

Numerical Exercise #1

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November 15, 2025

1 Question #1

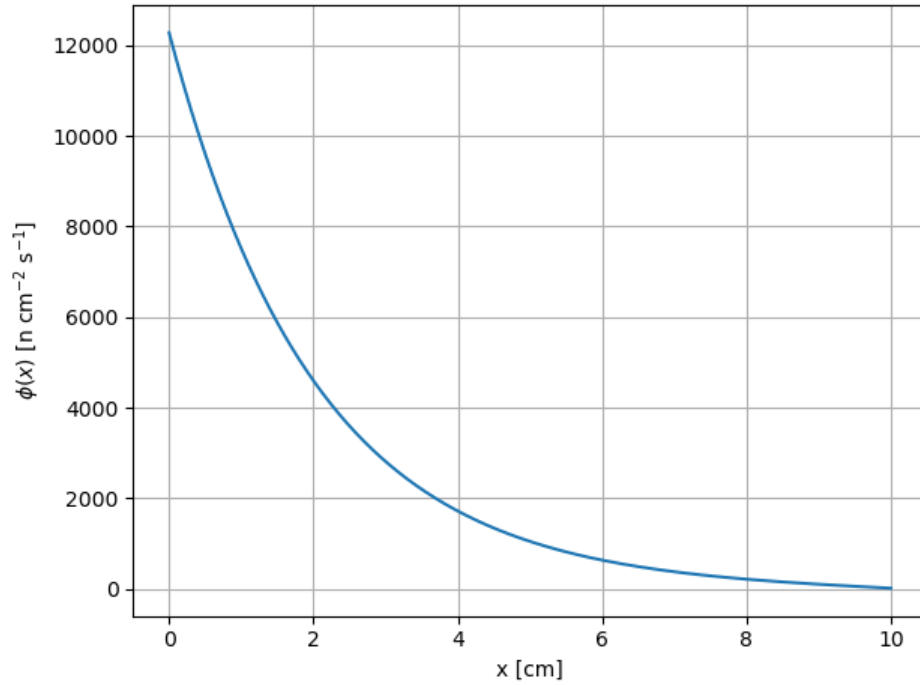


Figure 1: Analytical Solution for the neutron flux as a function of depth.

Flux values at the following locations (scientific format with 4 decimal points):

Flux value at x_0 :

$$\Phi(x_0) = 1.4405 \times 10^1 \text{ n cm}^{-2} \text{ s}^{-1}$$

Flux value at 0:

$$\Phi(0) = 1.2277 \times 10^4 \text{ n cm}^{-2} \text{ s}^{-1}$$

2 Question #2

Relationship between Φ_i , Φ_{i+1} , Φ_{i-1} at any point within the material:

$$A\Phi_i + B\Phi_{i-1} + C\Phi_{i+1} = D \quad (1)$$

Coefficients of the matrix A (with 4 decimal points), for a mesh size of 0.1 cm:

Coef $A_{i,i}$: 16.6037 cm^{-1}

Coef $A_{i-1,i}$: -8.2919 cm^{-1}

Coef $A_{i+1,i}$: -8.2919 cm^{-1}

Relationship between Φ_i , Φ_{i-1} , Φ_{i+1} ,
at the source:

$$A\Phi_1 + B\Phi_2 = C \quad (2)$$

at the RHS of the problem:

$$A\Phi_n + B\Phi_{n-1} = C \quad (3)$$

Associated coefficients of the matrix A (with 4 decimal points), for a mesh size of 0.1 cm:

at the source:

Coef $A_{1,i}$: 8.3119 cm^{-1}

Coef $A_{2,i}$: -8.2919 cm^{-1}

at the RHS:

Coef $A_{n-1,i}$: -8.2919 cm^{-1}

Coef $A_{n,i}$: 12.1536 cm^{-1}

3 Question #3

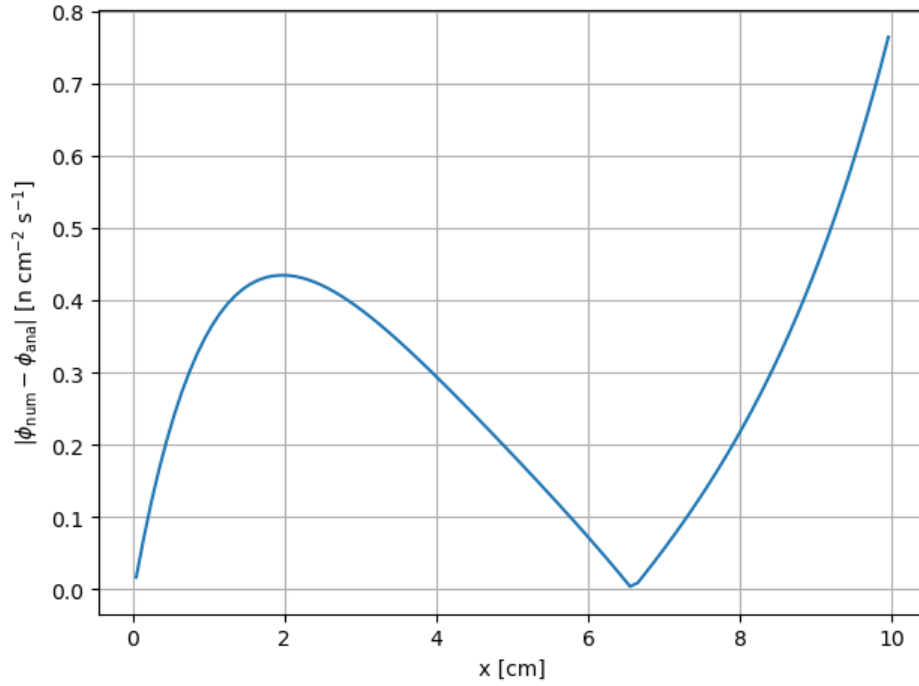


Figure 2: Distance between the solutions at each mesh point for a mesh size of 0.1 cm.

Flux values from the numerical solver at the following locations (scientific format with 4 decimal points):

Flux value at x_0 :

$$\Phi_{\text{num}}(x_0) = 1.3624 \times 10^1 \text{ n cm}^{-2} \text{ s}^{-1}$$

Flux value at 0:

$$\Phi_{\text{num}}(0) = 1.2281 \times 10^4 \text{ n cm}^{-2} \text{ s}^{-1}$$

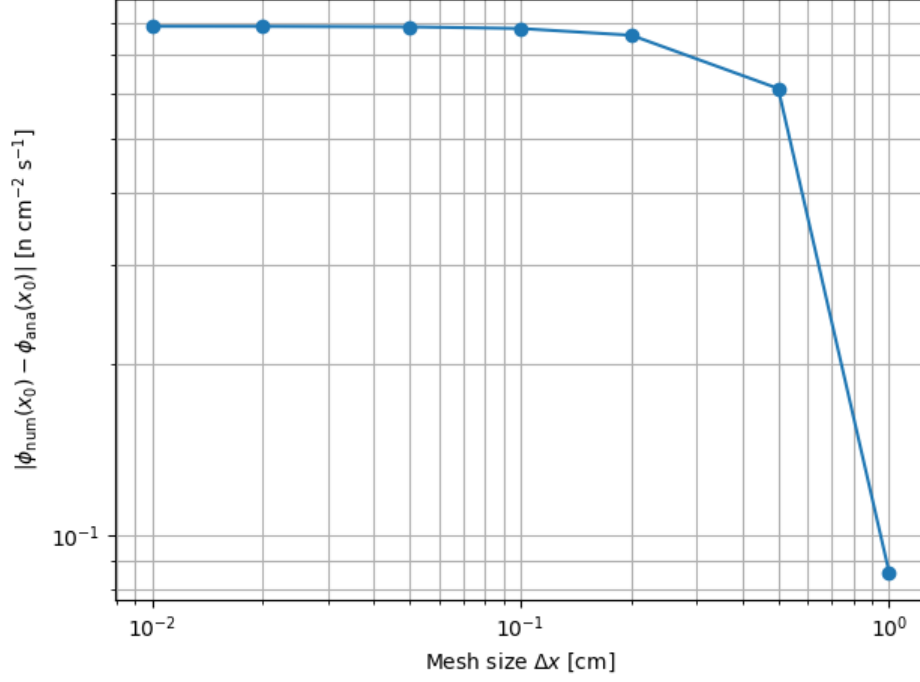


Figure 3: Evolution with mesh size of the absolute error of $\Phi(x_0)$.

The absolute error at $\phi(x_0)$ does not decrease monotonically with mesh refinement; it stays nearly constant for small Δx and changes mainly for the coarsest meshes. This indicates the error is dominated by the approximate vacuum boundary treatment rather than by the second-order interior finite-difference discretization.